METRIC

DOD-I-63274A
14 February 1983
SUPERSEDING
DOD-I-63274
26 September 1977

# MILITARY SPECIFICATION

## INSERT, SCREW THREAD - THIN WALL, LOCKED IN, METRIC, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers metric thin wall, locked in, screw thread inserts with and without metallic internal thread locking feature.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| QQ-A-225/6 | - | Aluminum Alloy Bar, Rod and Wire; Rolled, Drawn or Cold Finished, 2024 |
| QQ-A-250/6 | - | Aluminum Alloy 5083, Plate and Sheet |
| QQ-P-416 | - | Plating, Cadmium (Electrodeposited) |
| PPP-H-1581 | - | Hardware (Fasteners and Related Items), Packaging Of |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TST-S, Dover, NJ 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5340

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

MILITARY

| | | |
|---|---|---|
| MIL-I-6866 | - | Inspection, Penetrant Method Of |
| MIL-I-6868 | - | Inspection Process, Magnetic Particle |
| MIL-A-8625 | - | Anodic Coatings, for Aluminum and Aluminum Alloys |
| MIL-L-8937 | - | Lubricant, Solid Film, Heat Cured, Corrosion Inhibiting |
| MIL-I-63274/1 | - | Insert, Screw Thread-Thin Wall, Locked In, Metric |

STANDARDS

FEDERAL

| | | |
|---|---|---|
| FED-STD-H28/20 | - | Inspection Methods for Acceptability of UN, UNR, UNJ, M and MJ Screw - Threads |
| FED-STD-H28/21 | - | Screw Thread Standards for Federal Services, Section 21, Metric Screw Threads |

MILITARY

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-1312 | - | Fasteners Test Methods |

(Copies of specifications, standards, handbooks, drawings, and publication required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

| | | |
|---|---|---|
| ANSI B46.1 | - | Surface Texture |

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AEROSPACE MATERIAL SPECIFICATIONS (AMS)

| | | |
|---|---|---|
| AMS 2411 | - | Silver Plating For High Temperature Applications |
| AMS 5643 | - | Steel Bars, Forgings, Tubing and Rings, Corrosion Resistant |
| AMS 5731 | - | Steel Bars, Forgings, Tubing & Rings, Corrosion and Heat Resistant |
| AMS 5734 | - | Steel Bars, Forgings, and Tubing, Corrosion and Heat Resistant |
| AMS 6370 | - | Steel Bars, Forgings and Rings |

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 380 - Cleaning and Descaling Stainless Steel Parts Equipment and Systems.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Material. Recycled and reclaimed materials (See 3.1) shall be used to the maximum extent practicable.

3.2.1 Alloy steel. Alloy steel shall be composition 4130 (UNS G41300) in accordance with AMS 6370. Maximum rated temperature is 235°C for alloy steel inserts.

3.2.2 Corrosion-resistant steel. Corrosion resistant steel shall be composition 17-4PH (UNS S17400) in accordance with AMS 5643 or composition A-286 (UNS K66286) in accordance with AMS 5731 or AMS 5734. Maximum rated temperature for 17-4PH inserts is 310°C and for A-286 inserts is 650°C.

3.3 Treatment and finish.

3.3.1 Protective finish. The insert shall be furnished with a protective finish or surface treatment as specified.

3.3.1.1 Alloy steel inserts. Alloy steel inserts shall be cadmium plated in accordance with QQ-P-416, Type II, class 3. The plating thickness shall be 5.1 µm (micrometers). In addition, the alloy steel inserts shall be coated with a solid film lubricant in accordance with MIL-L-8937.

3.3.1.2 Corrosion-resistant steel inserts. Corrosion-resistant steel inserts shall be cleaned and descaled in accordance with ASTM A380. Inserts made from 17-4PH steel shall be coated with a solid film lubricant in accordance with MIL-L-8937. Inserts made from A-286 steel shall be silver plated in accordance with AMS 2411, 5µm thick minimum.

3.4 Design, dimensions and tolerances. Design, dimensions, and tolerances shall conform to the requirements of the applicable specification sheet.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

3.4.1 Threads. Threads shall be right hand, M or MJ profile in accordance with FED-STD-H28/21 and the applicable specification sheet.

3.4.1.1 External threads. External threads shall be class 4g, except minor diameters shall be in accordance with the applicable specification sheet. Thread gaging shall be performed after plating and prior to the application of solid film lubricant.

3.4.1.1.1 Lead threads. External lead threads shall not exceed two pitches, including chamfer.

3.4.1.2 Internal threads. Internal threads of alloy steel inserts and 17-4PH steel inserts shall be class-4H6H for thread sizes through MJ5X0.8 and class-4H5H for thread sizes greater than MJ5X0.8. Internal threads of A-286 steel inserts shall be class-5G. Thread gaging shall be performed after plating and prior to the application of solid film lubricant. For inserts with internal thread locking feature, the "GO" plug gage shall enter a minimum of 3/4 turn before engagement of the locking feature.

3.4.1.3 Thread forming. Threads shall be produced either by machining, grinding or fully formed by a single rolling process prior to protective plating or surface treatment.

3.4.1.4 Concentricity. The internal thread pitch diameter shall be concentric with the external thread pitch diameter within 0.15mm FIM.

3.4.2 Internal thread locking feature. The middle portion of the internal thread of the insert shall be distorted to impose friction between the insert and the inserted item. The self-locking insert shall be characterized by three axial grooves in the internal thread. The feature shall not operate by means of separate movement from the installation, nor depend upon axial load on the inserted item, but shall be an integral part of the insert wall.

3.4.2.1 Self-locking torque. When the insert is properly installed in the parent material, the locking feature shall provide a locking torque with a mating screw or bolt that is within the limits specified in Table I when tested in accordance with 4.7.2. Inserts shall not be subjected to self-locking tests after testing as specified in 4.7.3, 4.7.4 or 4.7.5.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

TABLE I. Self-locking torque

| Internal Thread Size | Maximum Locking Torque N·m | Minimum Breakaway Torque N·m |
|---|---|---|
| MJ3X0.5 | 0.6 | 0.09 |
| MJ4X0.7 | 0.9 | 0.14 |
| MJ5X0.8 | 1.6 | 0.23 |
| MJ6X1 | 2.8 | 0.36 |
| MJ8X1.25 | 5.8 | 0.71 |
| MJ10X1.5 | 9.6 | 1.16 |
| MJ12X1.25 | 14.8 | 1.77 |
| MJ14X1.5 | 22.6 | 2.71 |

3.5 Surface texture. The surface texture of the insert, prior to plating, shall not exceed the values specified in the applicable specification sheet and shall be in accordance with ANSI B46.1.

3.6 Metallurgical properties.

3.6.1 Cracks. (See 6.3.1) The insert shall be free of cracks in any direction or location.

3.6.2 Discontinuities.

3.6.2.1 Laps and seams. (See 6.3.2 and 6.3.3) Insert threads shall show no evidence of laps or seams at the root or along the flanks inside the pitch diameter. Laps and seams which do not exceed 20 percent of basic thread height are permissible at the crest and on the flanks outside the pitch diameter. Slight deviation from thread contour is permissible at the crest. The incomplete thread at each end of the insert may also deviate slightly from contour.

3.6.2.2 Inclusions. (See 6.3.4) The insert shall show no evidence of surface or subsurface inclusions at the thread root. Small inclusions in other parts of the insert not indicative of unsatisfactory quality shall be permissible.

3.7 Mechanical properties.

3.7.1 Hardness. The insert shall be within the hardness range as specified below when tested in accordance with 4.7.1.

| | | |
|---|---|---|
| Alloy steel | - | 24-34 HRC |
| 17-4PH steel | - | 35-42 HRC |
| A-286 steel | - | 32-40 HRC |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

3.7.2 Tensile strength. The properly installed insert shall be capable of withstanding the tensile strengths specified in Table II without evidence of thread damage or failure when tested in accordance with 4.7.3. Tensile strengths are based on the tensile stress areas of the test bolts or screws and a minimum ultimate tensile strength of 1100 MPa.

TABLE II. Tensile strength-internal thread

| Internal Thread Size | Tensile Stress Area $mm^2$ | Minimum Tensile Strength kN |
|---|---|---|
| MJ3X0.5 | 4.96 | 5.46 |
| MJ4X0.7 | 8.64 | 9.5 |
| MJ5X0.8 | 14.0 | 15.4 |
| MJ6X1 | 19.8 | 21.8 |
| MJ8X1.25 | 36.1 | 39.7 |
| MJ10X1.5 | 57.3 | 63 |
| MJ12X1.25 | 91.3 | 100 |
| MJ14X1.5 | 123.5 | 136 |

3.7.3 Resistance to pullout. The insert shall demonstrate minimum shear engagement areas as specified in Table III when tested in accordance with 4.7.4. The properly installed insert shall have a minimum resistance to pullout from the parent material as specified in Table III. Values specified in Table III are based on the minimum shear engagement areas of the insert when installed in a test block having an ultimate shear strength of 170MPa.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

TABLE III. Shear engagement area and resistance to pullout - external thread

| Internal Thread Size | D63274/1 Dash Number | Minimum Shear Engagement Area $mm^2$ | Minimum Resistance to Pullout kN |
|---|---|---|---|
| MJ3X0.5 | 1, 2 | 27.78 | 4.72 |
| MJ4X0.7 | 3, 4 | 54.46 | 9.26 |
| MJ5X0.8 | 5, 6 | 69.39 | 11.8 |
| MJ5X0.8 | 7, 8 | 95.74 | 16.3 |
| MJ6X1 | 9,10 | 104.48 | 17.8 |
| MJ6X1 | 11,12 | 140.71 | 23.9 |
| MJ8X1.25 | 13,14 | 217.61 | 37 |
| MJ8X1.25 | 15,16 | 290.82 | 49.5 |
| MJ10X1.5 | 17,18 | 352.72 | 60 |
| MJ10X1.5 | 19,20 | 464.64 | 79 |
| MJ12X1.25 | 21,22 | 525.64 | 89 |
| MJ12X1.25 | 23,24 | 695.43 | 118 |
| MJ14X1.5 | 25,26 | 724.49 | 123 |
| MJ14X1.5 | 27,28 | 950.53 | 162 |

NOTES:

1. Shear engagement area is the assembled dimensional value for the overall engaged area of mating thread members. It does not represent a dimension of either of the members in an unassembled condition.

2. Pullout load = Shear engagement area x 170 MPa. To compute pullout load in other materials, multiply Shear engagement area by applicable ultimate shear strength of material.

3.7.4 Rotational resistance. The insert when properly installed shall withstand the torque values specified in Table IV without failure when tested in accordance with 4.7.5.

TABLE IV. External rotational resistance

| Internal Thread Size | Minimum Rotational Resistance N•m | Internal Thread Size | Minimum Rotational Resistance N•m |
|---|---|---|---|
| MJ3X0.5 | 1.8 | MJ8X1.25 | 17.4 |
| MJ4X0.7 | 2.7 | MJ10X1.5 | 28.8 |
| MJ5X0.8 | 4.8 | MJ12X1.25 | 44.4 |
| MJ6X1 | 8.4 | MJ14X1.5 | 67.8 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

3.8 Workmanship. Workmanship shall be consistent with the type of product, finish and class of thread fit specified. The insert shall be of uniform quality and free from defects which would be detrimental to the performance of the insert in service use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. The manufacturer shall insure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.1.2 Test block fabrication. Test block shall be fabricated in accordance with figure 1. Test block may exceed dimensions specified in figure 1 to accommodate multiple testings of inserts except for tests of 4.7.3. and 4.7.4.

4.1.3 Test specimen installation. Sample inserts shall be installed in accordance with installation methods specified in the applicable specification sheet in test block specified in 4.1.2. Test specimens are not required for 4.6.2, 4.6.4, and 4.7.1.

4.1.3.1 Inspection of installed inserts. The inserts installed as specified in 4.1.3 shall be visually inspected under 10 diameters magnification. The presence of cracks in either test block or insert as a result of installation shall be cause for rejection. When visual inspection discloses a condition which shows cause for further examination, the test specimen shall be penetrant inspected in accordance with MIL-I-6866.

4.1.3.2 Test bolts and screws. Bolts and screws for use in all tests shall have class 4h threads in accordance with FED-STD-H28/21.

4.1.3.2.1 Bolts and screws used for tests of 4.7.3, 4.7.4 and 4.7.5 shall be heat treated to 1250 MPa minimum ultimate tensile strength and shall be cadmium plated in accordance with QQ-P-416, Type II, Class 3, (5.1 µm plating thickness).

4.1.3.2.2 Bolts and screws used for test of 4.7.2 shall be corrosion resistant steel, cleaned and descaled in accordance with ASTM A380 for testing corrosion resistant steel inserts. Bolts and Screws used for testing alloy steel inserts shall be non-corrosion resistant steel, cadmium plated.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

4.2 Quality conformance inspection. Quality conformance inspection shall be as specified in table V.

4.2.1 Inspection lot. An inspection lot shall consist of all inserts covered by a single specification sheet produced under essentially the same conditions and offered for inspections at one time.

4.2.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. A resubmitted lot shall be inspected using tightened inspection. Such lots shall be separate from new lots and clearly identified as reinspected lots.

4.3 Classification of defects. Classification of defects shall be in accordance with MIL-STD-105 and table VI.

4.3.1 Defect noncompliance. An insert exhibiting one or more defects shall be considered defective.

4.4 Inspection sampling. Inspection sampling shall be in accordance with MIL-STD-105 and the applicable inspection level and Acceptable Quality Level (AQL) specified in table VI.

4.5 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with requirements of PPP-H-1581.

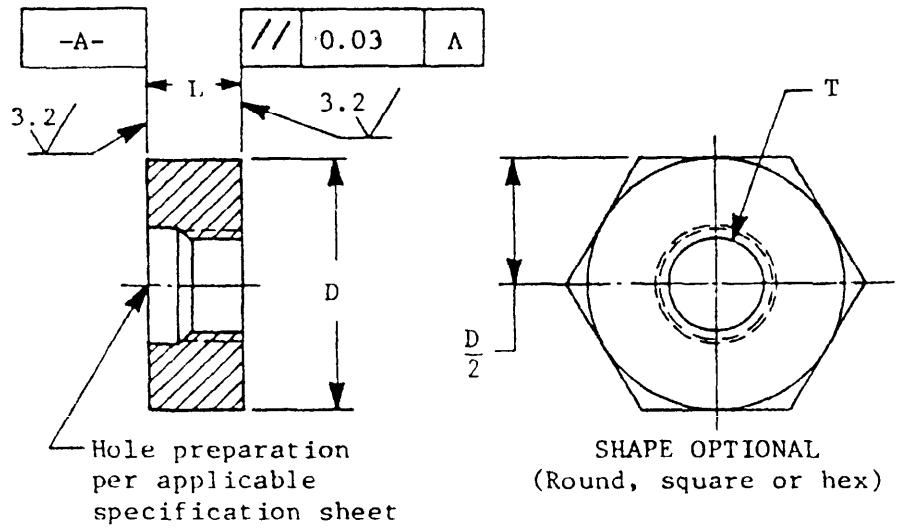
4.6 Methods of inspection.

4.6.1 Visual and dimensional. The insert shall be examined to verify that physical dimensions, surface texture, and workmanship are in accordance with the applicable requirements of 3.4, 3.5, and 3.8.

4.6.2 Material inspection. Material inspection shall consist of certification supporting verifying data that the materials used in fabricating the insert are in accordance with the applicable requirements of 3.2.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.




NOTES:

1. DIMENSIONS:

   T = Nominal diameter of insert external thread.
   D = 4 x T
(a) L = Insert length minus 1.6mm for rotational resistance test (4.7.5).
(b) L = Insert length plus 1.6mm for self-locking torque (4.7.2) tensile strength (4.7.3) and resistance to pullout (4.7.4) tests.

2. MATERIAL:

(a) Self-locking torque (4.7.2), tensile strength (4.7.3) and rotational resistance (4.7.5) tests - Aluminum alloy, 2024-T4 in accordance with QQ-A-225/6.
(b) Resistance to pullout test (4.7.4) - Aluminum alloy, 5083-H321 in accordance with QQ-A-250/6.

3. Anodize in accordance with MIL-A-8625, Type I, Class 1 or 2.

4. Penetrant inspect in accordance with MIL-I-6866. Discontinuities or cracks shall not be acceptable.

FIGURE 1. Test block.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

TABLE V. Quality conformance inspection

| Inspection | Requirement Paragraph | Test Method Paragraph |
|---|---|---|
| Group A | | |
| Visual & Dimensional | 3.1 | 4.6.1 |
| Material | 3.2 | 4.6.2 |
| Treatment and Finish | 3.3(as applicable) | 4.6.3(as applicable) |
| Workmanship | 3.8 | 4.6.1 |
| Packaging | 5.1 | 4.5 |
| Group B | | |
| Threads | 3.4.1 | 4.6.4 |
| Concentricity | 3.4.1.4 | 4.6.4.3 |
| Internal thread locking feature | 3.4.2 | 4.7.2<br>4.6.4.4 |
| Surface texture | 3.5 | |
| Group C | | |
| Self-locking torque | 3.4.2.1 | 4.7.2.1 |
| Minimum breakaway torque | 3.4.2.1 | 4.7.2.2 |
| Cracks | 3.6.1 | 4.6.4.5 |
| Laps and seams | 3.6.2.1 | 4.6.4.5 |
| Inclusions | 3.6.2.2 | 4.6.4.5 |
| Hardness | 3.7.1 | 4.7.1 |
| Tensile strength | 3.7.2 | 4.7.3 |
| Resistance to pullout | 3.7.3 | 4.7.4 |
| Rotational resistance | 3.7.4 | 4.7.5 |



Source: https://assist.dla.mil -- Downloaded: 202610-26T04:26Z
Check the source to verify that this is the current version before use.

TABLE VI. Classification of defects & inspection sampling

| Category | Defect | Inspection Method |
|---|---|---|
| Critical | None defined | |
| Major | AQL = 1.5 percent defective, Level II | |
| 101 | Design and dimensions incorrect (3.4) | Standard Inspection Equipment (SIE) |
| 102 | Threads not as specified (3.4.1) | SIE |
| 103 | Concentricity (3.4.1.4) | SIE |
| 104 | Self-locking feature missing, when required (3.4.2) | Visual |
| 105 | Surface texture (3.5) | SIE |
| 106 | Cracks (3.6.1) and Discontinuities (3.6.2) | SIE |
| Minor | AQL = 2.5 percent defective, Level S1 | |
| 201 | Treatment and finish (3.3 as applicable) | Visual |
| 202 | Lead threads (3.4.1.1.1) | Visual |
| 203 | Workmanship (3.8) | Visual |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

4.6.3 Treatment and finish inspection.

4.6.3.1 Protective finish inspection. Sample inserts shall be inspected for minimum thickness and continuity of plating in accordance with QQ-P-416 or AMS 2411 as applicable and the requirements of 3.3.1.

4.6.3.2 Cleaning and descaling inspection. Sample inserts shall be inspected for cleaning and descaling in accordance with ASTM A380 and 3.3.1.2.

4.6.4 Thread inspection.

4.6.4.1 External threads inspection. Sample inserts shall have their external threads inspected in accordance with System 21, FED-STD-H28/20, 3.4.1.1 and 3.4.1.3.

4.6.4.2 Internal threads inspection. Sample inserts shall have their internal threads inspected in accordance with System 21, FED-STD-H28/20, 3.4.1.2 and 3.4.1.3.

4.6.4.3 Concentricity inspection. Sample inserts shall have their internal thread pitch diameter concentric with the external thread pitch diameter in accordance with 3.4.1.4.

4.6.4.4 Surface texture inspection. Sample inserts shall be inspected for surface texture with any of the surface examination and measurement methods specified in ANSI B46.1 in accordance with the requirements of 3.5.

4.6.4.5 Metallurgical properties inspection. Sample inserts shall be inspected for cracks and discontinuities in accordance with 3.6. Any crack detected during visual examination shall be cause for rejection of the lot. When visual evidence of cracks or discontinuities show cause for further inspection, sample inserts shall be subjected to magnetic particle inspection in accordance with MIL-I-6868 for alloy steel and penetrant inspection in accordance with MIL-I-6866 for corrosion resistant steel. Magnetic particle or penetrant inspection alone shall not be cause for rejection. If indications are considered cause for rejection, representative samples shall be taken from those inserts showing indications and these samples shall be further examined. Inserts shall be sectioned and cracks and discontinuities measured under 10X magnification. The inspection shall be performed on finished inserts, free of lubrication and subsequent to any processing operation which could adversely affect the inserts. Requirements for dye as an indication of particle inspection may be waived.

4.7 Mechanical properties inspection.

4.7.1 Hardness test. Sample inserts shall be tested for hardness to meet the requirements of 3.7.1 in accordance with MIL-STD-1312, test 6.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

4.7.2 Self-Locking torque test. Sample inserts shall be installed in test blocks in accordance with figure 1 and 4.1.3 to meet the requirements of 3.4.2.1. Bolts and screws shall be in accordance with 4.1.3.2 and shall have sufficient thread length to extend beyond the locking feature a minimum of two pitches (including thread chamfer). A new bolt or screw and a new sample insert shall be used for each complete fifteen cycle test. Bolts and screws shall assemble freely, with the fingers, up to the locking feature. The bolt or screw shall be engaged or disengaged from the assembled insert self-locking area for fifteen full installation and removal cycles without axial load on the insert. The test shall be run at a rate slow enough to yield a dependable measure of torque and to avoid heating of the bolt or screw. A bolt or screw shall be considered fully installed when two threads extend past the end of the insert locking feature; the removal cycle shall be considered complete when the locking feature is disengaged.

4.7.2.1 Maximum locking torque test. Maximum locking torque shall be the maximum torque value encountered on any installation or removal cycle and shall not exceed the applicable values specified in Table I in accordance with the requirement of 3.4.2.1.

4.7.2.2 Minimum breakaway torque test. Minimum breakaway torque shall be the minimum torque required to start removal of the bolt or screw from a fixed position located between the first 1/4 turn to 1 turn of the removal cycle. Minimum breakaway torque readings shall be recorded at the start of the first, seventh, and fifteenth removal cycles. The torque for any cycle shall not be less than the applicable values specified in Table I in accordance with the requirements of 3.4.2.1.

4.7.3 Tensile strength test. Sample inserts shall be installed in test blocks in accordance with figure 1 and 4.1.3. The test bolt or screw shall assemble freely into the insert with finger torque prior to engaging the locking feature. The bolt or screw thread shall be of sufficient length to fully engage the entire length of the insert internal thread when the sample inserts are installed in test fixture in accordance with figure 2. The applicable tensile strength value specified in Table II shall be applied to the assembly and sample inserts shall meet the requirements of 3.7.2. In the event of bolt or screw failure below specified tensile strength value, the test shall be repeated until the tensile strength value of the insert is reached or exceeded. Rate of loading shall not exceed 700 MPa per minute based on the shank diameter area of the bolt or screw.

4.7.4 Resistance to pullout test. Sample inserts shall be installed in test blocks in accordance with figure 1 and 4.1.3. The minimum single shear strength of the test block material shall be determined by means of double shear tests on coupons taken from the material from which the test blocks are fabricated. The procedure shall be the same as specified in 4.7.3 except, to demonstrate the average shear engagement area, an axial load of sufficient magnitude to produce failure (pullout) shall be applied to the assembly. The axial load test result shall be adjusted for test block shear strength as follows: The product of the load test results (N) and the ratio of the 170 MPa to the actual minimum shear



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

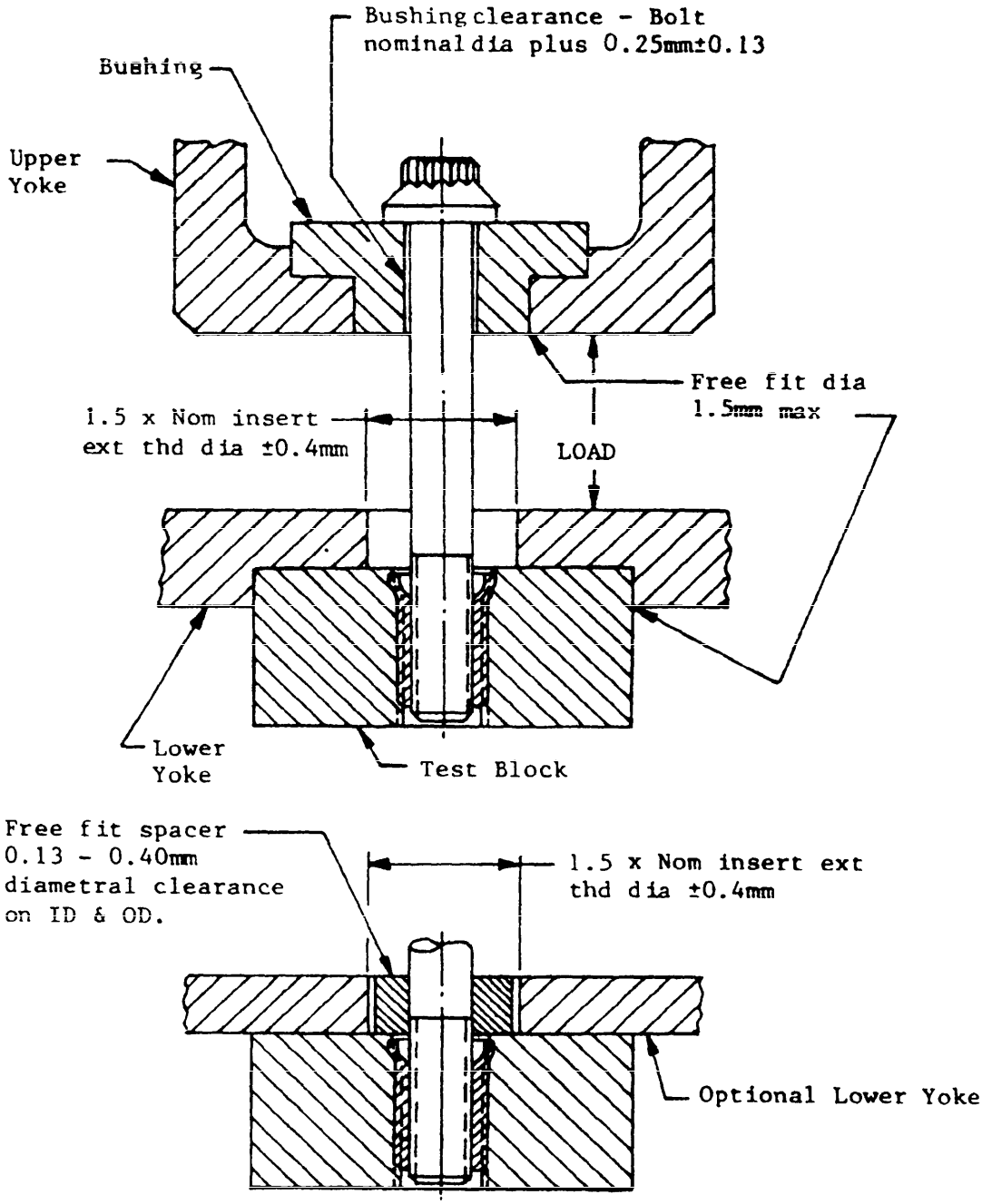
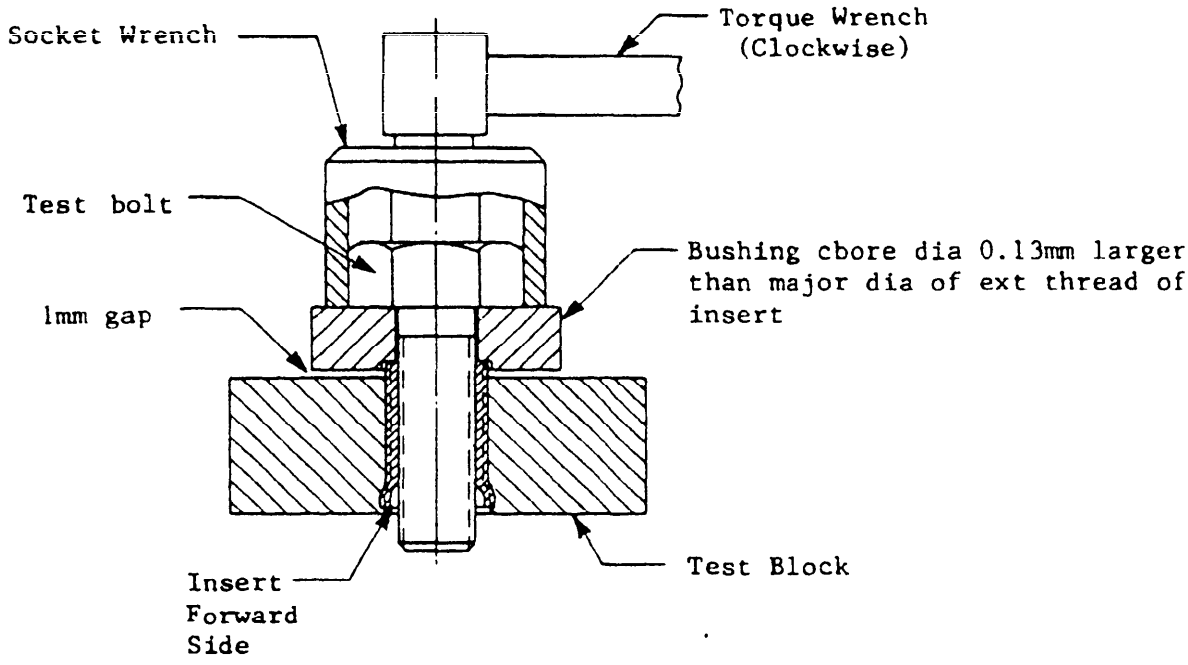


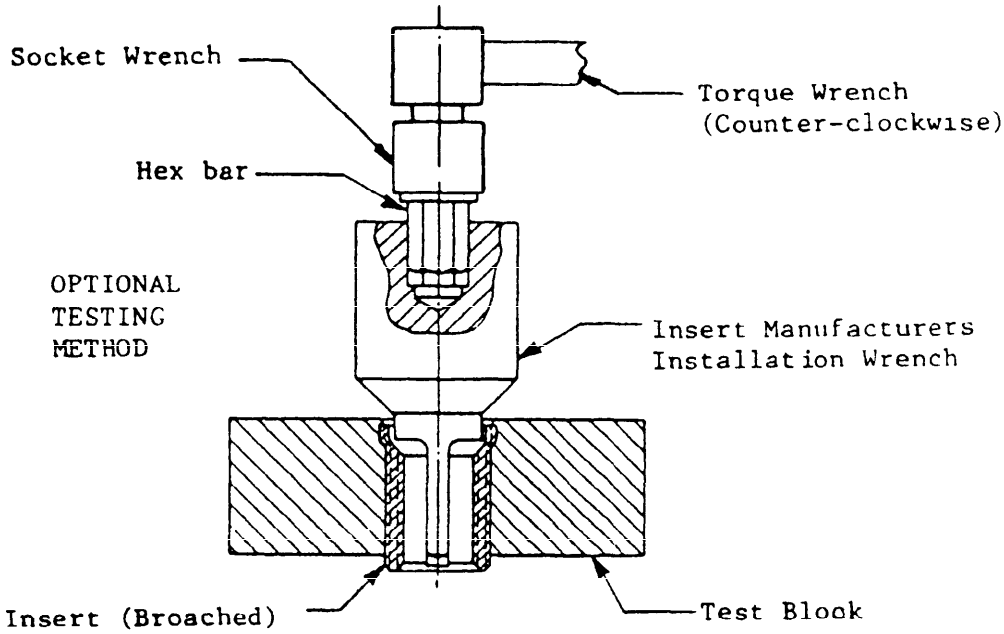

FIGURE 2. Tensile strength and resistance to pullout test fixtures.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.




NOTE: Above test fixture used with insert in the as received condition.




NOTE: Above test fixture used when product is internally broached with a two or three-slot recess whose cross point dimension does not exceed the major diameter of the insert internal thread. More accurate results are obtainable with this method as the insert will not tend to collapse or transmit friction into final values.

FIGURE 3. Rotational resistance test fixture



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

strength shall be used as the resistance to pullout (N). The mean (arithmetical average) value, as determined by five or more tests, shall meet or exceed the values specified in Table III in accordance with 3.7.3. In the event of bolt or screw failure below the applicable strength rating, the test shall be repeated until the pullout strength of the insert is reached or exceeded. Rate of loading shall not exceed 700 MPa per minute based on the shank diameter area of the bolt or screw. Whenever the pullout values exceed the tensile strength of the test bolt or screw, or the test loads exceed the capabilities of the test equipment, it is permissable to use lower shear strength materials (100 MPa approx.) in lieu of specified test block materials. Double shear tests on coupons taken from the lower shear strength test blocks shall be performed and the resultant shear strength used to adjust the test results as stated above.

4.7.5 Rotational resistance test. Sample inserts shall be installed in test blocks in accordance with figure 1 and 4.1.3 and utilizing a test assembly as illustrated in figure 3. With the forward side face of the insert installed in accordance with 4.1.3, the far side of the insert shall extend a distance approximately 1.6mm past the far side of the test block allowing it to fit into the counterbore of the bushing. The insert shall fit into the bushing counterbore sufficiently close to prevent it from expanding under pressure. A test bolt or screw shall enter the insert from the back side (opposite normal entry). The test bolt or screw shall be torqued in a clockwise direction. Failure at applicable values below those specified in Table IV in accordance with the requirements of 3.7.4 shall be cause for rejection.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with PPP-H-1581 (See 6.2).

6. NOTES

6.1 Intended use. Inserts covered by this specification are intended for use as a general purpose, locked in, metric fastener, with or without an internal metallic locking feature to resist rotation of a bolt or screw.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification and the applicable specification sheet.

b. Applicable specification sheet part number (See 3.1).

c. Level (degree) of protection, in accordance with PPP-H-1581, ordering data (See 5.1).

6.3 Definitions.

6.3.1 Crack. A crack is defined as a clean crystalline break passing through the grain or grain boundary without the inclusion of foreign elements.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

6.3.2 Lap. A surface defect appearing as a seam caused by folding over metal, fins or sharp corners, and then rolling or forging, but not welding them into surface.

6.3.3 Seam. An elongated discontinuity caused by a defect, which has been closed by rolling or forging mechanically, but not united.

6.3.4 Inclusions. Particles of nonmetallic impurities such as oxides, sulfides and silicates, which are mechanically held in the steel during solidification.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to extensiveness of the changes.

Custodians:
Army - AR
Navy - OS
Air Force - 99

Review activities:
Army - AV, MI
Air Force - 11
DLA - IS

User activities:
Army - ME, AT, CR
Navy - MC

Preparing activity:
Army - AR

(Project 5340-1461)



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS This form is provided to solicit beneficial comments which may improve this document and enhance its use DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed

NOTE This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

DOCUMENT IDENTIFIER *(Number)* AND TITLE

DOD-I-63274A INSERT, SCREW THREAD-THIN WALL, LOCKEDIN, METRIC,

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDED WORDING CHANGE

C REASON FOR RECOMMENDED CHANGE(S)

2 REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO

DATE

**DD FORM 1426** 1 OCT 76

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.